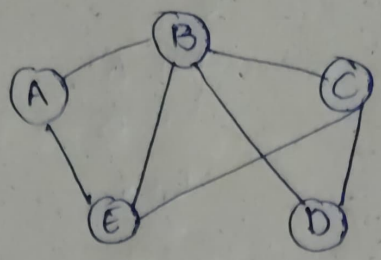
# Tut-7

1

G =




a) adjacency matrix

| | A | B | C | D | E |
|---|---|---|---|---|---|
| A | 0 | 1 | 0 | 0 | 1 |
| B | 1 | 0 | 1 | 1 | 1 |
| C | 0 | 1 | 0 | 1 | 1 |
| D | 0 | 1 | 1 | 0 | 0 |
| E | 1 | 1 | 1 | 0 | 0 |

b) adjacency-list

A → B, E

B → A, C, D, E

C → B, D, E

D → B, C

E → A, B, C

c) bipartile :- it has no odd cycle

Ans, B-C-E-B is the odd triangle(cycle)

d) connected

Yes, Every vertex can be reachead from any other vertex.

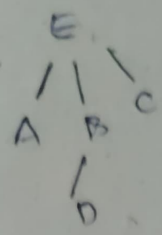
e) simple : no self loops & no multiple edges

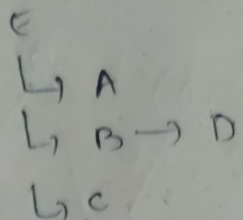
Yes,

f) d(A, D)

d(A, B) : 2 (min)

2) BFS from E




3) $G = (V, E)$ $V$ = no. of vertices, $E$ = no. of edges

| operation | adjacency matrix | adjacency list |
|---|---|---|
| a) test if $uv \in E$ | $O(\deg(u))$ | $O(1)$ |
| b) list $v$'s out-neighbours | $O(\deg(v))$ | $O(V)$ |
| c) insert edge $uv$ | $O(1)$ | $O(1)$ |
| d) delete edge $uv$ | $O(\deg(u))$ | $O(1)$ |
| e) list all edges | $O(V+E)$ | $O(V^2)$ |

4) A graph $G$ is bipartite, iff $G$ has no odd length cycle.

Proof:

If $G$ is bipartite then it has no odd cycle.

let the bipartition be $V = V_1 \cup V_2$

Every edge connects a vertex in $V_1$ to a vertex in $V_2$ in any cycle.

Vertices must alternate b/w $V_1$ & $V_2$

so cycle must look like

$V_1 \rightarrow V_2 \rightarrow V_1 \rightarrow V_2 \cdots$

To return to the starting vertex the no of edges must be even

∴ odd length cycles cannot exist

⇒ if G has no odd cycle then it is bipartite

choose any vertex & perform BFS

Divide vertices into 2 sets based on level:

$V_1$: even level vertices $V_2$: odd level vertices

If an edge connects 2 vertices in the same set

it creates an odd cycle.

But the graph has no odd cycle.

Thus such edges cannot exist

∴ G is bipartite (⇐) G has no odd cycle

5) $G = (V, E)$, $G^T = (V, E^T)$

$E^T = \{(v, u) \mid (u, v) \in E(G)\}$ all edges are reversed.

*computing $G^T$ from G

i) using adjacency list

for each vertex u in V:

for each v in adj[u]:

add edge (v, u) to adjT[v]

TC = O(V+E) (each vertex & edge is processed once)

ii) using adjacency matrix.

AT[i][j] = A[j][i]

for i = 1 to V

for j = 1 to V

AT[i][j] = A[j][i]

TC = O(V²)

6) $G^2 = (V, E^2)$ s.t $(u, v) \in E(G^2)$

if ∃ a path from u to v of length ≤ 2 direct edge (or through one immediate vertex

Q) G ⇒ A → B, E,

B → A, E, C, D

C → B, D, E

D → B, C

E → A, B, C

A → C (via B or E)

A → D (via B)

B → A, B → B, B → D, D → E already done

E → E (via B or D)

$G^2$ =)

| | |
|---|---|
| A-B | B-B |
| A-C | B-E |
| A-D | C-D |
| A-E | C-E |
| B-C | D-E |

*almost complete graph except maybe some duplicates

b) Adjacency list

```
for each vertex u
    for each v in adj[u]
        for each w in adj[v]
            add edge(u,w)
```

$T(n) = O(V \times d^2)$

d: degree

$T(n) = O(V^3)$ (worst)

Adjacency matrix

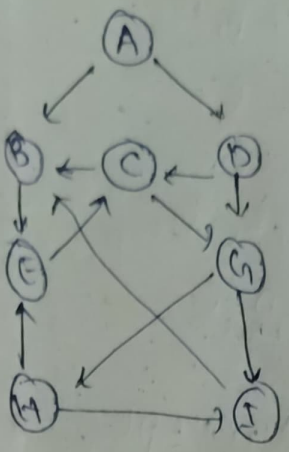
compute $G^2 = A^2$

matrix multiplication

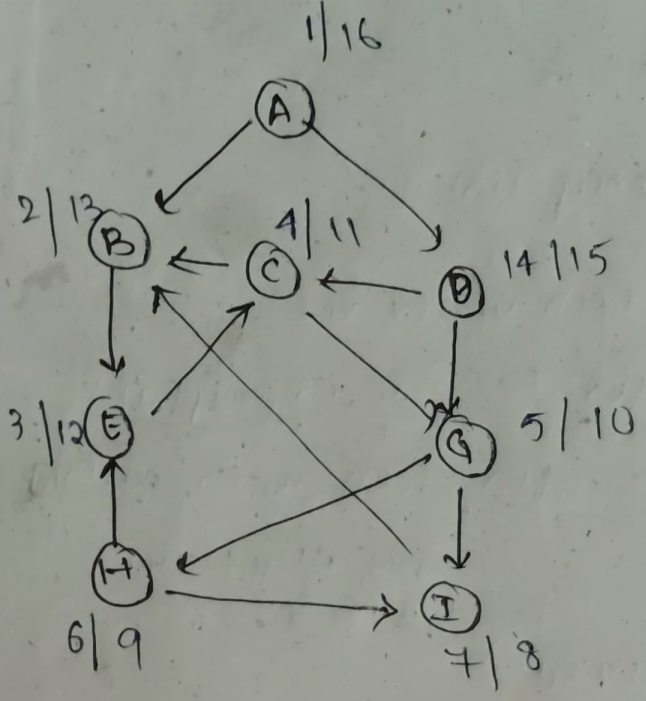
if $A^2[i][j] > 0$ then edge exists

$T(n) = O(V^3)$

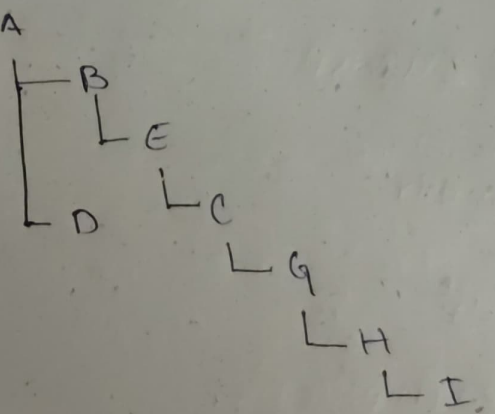
4)




a)




b)




c) Tree edges:

A→B, B→E, E→C, C→G, G→H, H→I, A→D

Back edges:

C→B, H→E, I→B

forward edges:

G → I

cross edges:

D → C, D → F

d) topological ordering doesnot exist.

because

B → E → C → B has a cycle

8) a) False - Graph with cycles does not have topological ordering.

b) False - A DAG may have multiple orderings

c) TRUE

d) TRUE.